

Rover Design Trade Study

Compare three rover concepts using mission-weighted criteria and recommend one for detailed design.

Advanced	2 class periods	Engineering Challenge
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Engineering Context

The project team has several promising rover concepts but limited time and budget. A structured trade study is required before committing to hardware.

What You Will Practice

- Select decision criteria
- Assign defensible weights
- Score concepts consistently
- Perform sensitivity analysis

What You Need

- Three concept descriptions or team sketches
- Decision matrix template
- Mission requirements
- Calculator or spreadsheet
- Engineering notebook

Student Instructions

- 1 Review the mission requirements and identify six to eight decision criteria.
- 2 Assign weights totaling 100%.
- 3 Define a 1-5 scoring scale for every criterion before scoring concepts.
- 4 Score all three concepts and document evidence.
- 5 Calculate weighted totals.
- 6 Change the two largest weights by plus/minus 10 percentage points and recalculate.
- 7 Identify whether the winner changes.
- 8 Recommend a concept and state the largest unresolved risk.

Engineering Requirements

- At least three concepts and six criteria
- Weights total 100%
- Scoring definitions written before final scoring
- All scores supported by evidence or assumptions
- Include one sensitivity analysis

Constraints & Safety

- Do not change criteria solely to make a preferred concept win
- Label uncertain assumptions
- Use the same scoring standard for all concepts

What You Submit

- Criteria and weights
- Scoring definitions
- Completed trade matrix
- Sensitivity comparison
- Recommendation memo

Reflection Questions

- 1 Which criterion drove the decision most strongly?
- 2 Did sensitivity analysis change confidence in the winner?
- 3 What missing test data would reduce uncertainty?
- 4 Where might personal bias have affected scoring?

Engineering Tip

Write scoring anchors such as “5 = meets requirement with at least 20% margin.”

Common Mistake

Assigning numbers without defined scoring rules creates a false appearance of precision.

Stretch Goal

Add cost and schedule uncertainty using best-case and worst-case scores.

Career Connection

Systems engineering teams use formal trade studies to select architectures, sensors, propulsion, and mission concepts.